

Exercice 1 :

```
function compl()
....j = %i
....z1 = -5.2*j - 2.5
....z2 = -3 - .4*j
....z3 = -3 + .4*j
....z4 = -1 - .j
....z5 = .4 + .2*j
....z6 = (-5 + .j) / (3 + 2*j)
....z7 = -1 - 2*j
....z8 = 5*j
....z9 = -7*j
....z10 = -3.93
....z11 = -2 + 2*j
....z12 = 5*j
....z13 = -7*j
....z14 = -3.93
....disp("partie réelle et imaginaire de " + string(z1), real(z1), imag(z1))
....disp("partie réelle et imaginaire de " + string(z2), real(%pi), imag(%pi))
....disp("partie réelle et imaginaire de " + string(z3), real(j), imag(j))
....disp(string(z2) + " + " + string(z3) + " = ", z2 + z3)
....disp(string(z2) + " - " + string(z3), z2 - z3)
....disp(string(z4) + " * " + string(z5), z4 * z5)
....disp(string(z2) + " * " + string(z3), z2 * z3)
....disp("partie réelle et imaginaire de " + string(z6) + " = ", real(z6), imag(z6))
....disp("module de " + string(z7) + " = ", abs(z7))
....disp("module de " + string(z8) + " = ", abs(z8))
....disp("module de " + string(z9) + " = ", abs(z9))
....disp("module de " + string(z10) + " = ", abs(z10))
....disp("argument de " + string(z11) + " = ", atan(imag(z11), real(z11)))
....disp("argument de " + string(z12) + " = ", atan(imag(z12), real(z12)))
....disp("argument de " + string(z13) + " = ", atan(imag(z13), real(z13)))
....disp("argument de " + string(z14) + " = ", atan(imag(z14), real(z14)))
....
endfunction
```

DIAZENZA MOUANDA Regis Japhet BALDE Mamadou

```
--> compl()

"partie réelle et imaginaire de -2.5+%i*5.2"

-2.5

5.2

"partie réelle et imaginaire de 3+%i*4"

3.1415927

0.

"partie réelle et imaginaire de 3+%i*4"

0.

1.

"3+%i*4 + 3+%i*4 = "

6. + 0.i

"3+%i*4 - 3+%i*4"

0. - 8.i

"1+%i * 4+%i*2"

6. - 2.i

"3+%i*4 * 3+%i*4"

25. + 0.i

"partie réelle et imaginaire de -1+%i = "

-1.

1.

"module de -1+%i*2 = "

2.2360680

"module de %i*5 = "

5.


"module de +%i*7 = "

7.

"module de -3.93 = "

3.93

2. + 2.i
```

Exercice 2 :

1) Forme algébrique à la main

$$z_1 + z_2 = 3+i + (-1+i) = 3+i-1+i = \underline{2+2i}$$

$$\begin{aligned} z_1 \times z_2 \times z_3 &= (3+i) \times (-1+i) \times (2+3i) \\ &= (-3+3i-i-1) \times (2+3i) = -4+2i \times (2+3i) = -8+6i-8+6i = \underline{-14+12i} \\ \frac{1}{z_1} &= \frac{1}{3+i} = \frac{3-i}{3^2+1^2} = \frac{3-i}{10} = \frac{3}{10} - \frac{1}{10}i = \underline{0,3-0,1i} \\ \frac{z_2}{z_3} &= \frac{-1+i}{2+3i} = \frac{(-1+i)(2-3i)}{2^2+3^2} = \frac{-2+3i+2i-3}{13} = \frac{-5+5i}{13} = \underline{-\frac{5}{13} + \frac{5}{13}i} \\ z_1^2 &= (3+i)^2 = (3+i)(3+i) = 9+3i+3i-1 = \underline{8+6i} \\ z_2^3 &= (-1+i)^3 = (-1+i)(-1+i)(-1+i) = (-1+i)(1-i) = (-1+i)(-1+i) = \underline{2-2i} \end{aligned}$$

$$\begin{aligned} \frac{1}{z_1} + \frac{1}{z_2} &= \frac{1}{3+i} + \frac{1}{2+3i} = \frac{1}{10} - \frac{1}{10}i + \frac{2-3i}{13} = \frac{3}{10} - \frac{1}{10}i + \frac{2-3i}{13} \\ &= \frac{3}{10} - \frac{1}{10}i + \frac{2}{13} - \frac{3}{13}i = \frac{3}{10} + \frac{2}{13} - \left(\frac{1}{10} + \frac{3}{13}\right)i \\ &= \frac{39+20}{130} - \frac{13+30}{130}i = \frac{59}{130} - \frac{43}{130}i = \frac{59-43i}{130} \\ &= \frac{130}{59-43i} = \frac{130(59+43i)}{59^2+43^2} = \frac{7670+5590i}{5330} \\ &= \frac{7670}{5330} + \frac{5590}{5330}i = \underline{1,4390 + 1,0487i} \end{aligned}$$

2) Vérification avec ScilaB

```
--> z1=3+%i;
--> z2=-1+%i
z2 =
-1. + i
--> z3=2+3*%i;
--> z1+z2
ans =
2. + 2.i
--> z1*z2*z3
ans =
-14. - 8.i
--> 1/z1
ans =
0.3 - 0.1i
--> z2/z3
ans =
0.0769231 + 0.3846154i
--> (z1)^2
ans =
8. + 6.i
--> (z2)^3
ans =
2. + 2.i
--> 1/((1/z1)+(1/z3))
ans =
1.4390244 + 1.0487805i
```

Exercice 3

1 – Calcul du module et de l'argument de z_1 et z_2

DIAZENZA MOUANDA Regis Japhet
BALDE Mamadou

```
> z1 = 1 + 3*i;  
> z2 = -2 + i;  
> disp("module de z1 = ",abs(z1),"module de z2 = ",abs(z2))  
  
"module de z1 = "  
  
3.1622777  
  
"module de z2 = "  
  
2.2360680  
> disp("argument en radian de z1 = ",atan(imag(z1),real(z1)),"argument en radian de z2 = ",atan(imag(z2),real(z2)))  
  
"argument en radian de z1 = "  
  
1.2490458  
  
"argument en radian de z2 = "  
  
2.6779450
```

Question 2 – 3

Exo 3

2. Déterminer les modules et arguments quand c'est possible

$$z_1 = 1 + 3i, |z_1| = 3 \times |z_1| = 3 \times 3,1622777 = 9,4868332$$

$$\frac{z_1}{z_2} = \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} = \frac{3,1622777}{2,2360680} = 1,414213566$$

$$z_1^2 = |z_1^2| = |z_1 \times z_1| = |z_1| \times |z_1| = 3,1622777 \times 3,1622777 = 10$$

$$\frac{1}{z_2} = \left| \frac{1}{z_2} \right| = \frac{1}{|z_2|} = \frac{1}{2,2360680} = 0,447213594$$

$$z_2^5 = |z_2^5| = |z_2| \times |z_2| \times |z_2| \times |z_2| \times |z_2| = 2,2360680^5 = 55,9017017$$

3. Mettre sous forme algébrique puis calculer le module et l'argument

$$z_1 + z_2 = 1 + 3i - 2 + i$$

$$z_1 + z_2 = -1 + 4i$$

$$| -1 + 4i | = \sqrt{(-1)^2 + (4)^2} = \sqrt{17}$$

$$\cos(\theta) = \frac{-1}{\sqrt{17}}$$

$$\sin(\theta) = \frac{4}{\sqrt{17}}$$

$$\theta = \arctan\left(\frac{4}{-1}\right) = 104,07^\circ$$

$$z_1^2 = (1 + 3i)^2$$

$$z_1^2 = 1 + 6i - 9$$

$$z_1^2 = -8 + 6i$$

$$|z_1^2| = \sqrt{(-8)^2 + (6)^2} = 10$$

$$\tan^{-1}\left(\frac{6}{-8}\right) = 143,13^\circ$$

$$z_2 = 3(1 + 3i)$$

$$z_2 = 3 + 9i$$

$$|3 + 9i| = \sqrt{3^2 + 9^2} = 3\sqrt{10}$$

$$\tan^{-1}\left(\frac{9}{3}\right) = 71,56^\circ$$

$$\frac{1}{z_2} = \frac{1}{-2 + i}$$

$$\frac{1}{z_2} = \frac{-2 - i}{(-2 + i)(-2 - i)} = \frac{-2 - i}{5}$$

$$\left| \frac{1}{z_2} \right| = \frac{\sqrt{5}}{5}$$

$$\tan^{-1}\left(\frac{-1}{-2}\right) = 110,7^\circ$$

$$z_1 + 2z_2 = 1 + 3i + 2(-2 - i) \times 3$$

$$z_1 + 2z_2 = 1 + 3i - 6 - 6i$$

$$z_1 + 2z_2 = -5 - 3i$$

$$|z_1 + 2z_2| = \sqrt{(-5)^2 + (-3)^2} = \sqrt{34}$$

$$\tan^{-1}\left(\frac{-3}{-5}\right) + 180^\circ = 129,6^\circ$$

$$z_2^5 = (-2 + i)^5$$

$$z_2^5 = (-2 + i)(-2 + i)(-2 + i)(-2 + i)(-2 + i)$$

$$z_2^5 = 38 + 41i$$

$$|z_2^5| = \sqrt{38^2 + 41^2} = 55,9017$$

$$\tan^{-1}\left(\frac{41}{38}\right) = 47,174^\circ$$

Question 4 : Vérification avec scilab

```
function compl()
... j = %i
... z1 = -1+3*j
... z2 = -2+j
... disp("z1+z2 = ", z1+z2, "|z1+z2| = ", abs(z1+z2), "arg(z1+z2) = ", atand(imag(z1+z2), real(z1+z2)))
... disp("3z1 = ", 3*z1, "|3z1| = ", abs(3*z1), "arg(3*z1) = ", atand(imag(3*z1), real(3*z1)))
... disp("z1/z2 = ", z1/z2, "|z1/z2| = ", abs(z1/z2), "arg(z1/z2) = ", atand(imag(z1/z2), real(z1/z2)))
... disp("z1^2 = ", z1^2, "|z1^2| = ", abs(z1^2), "arg(z1^2) = ", atand(imag(z1^2), real(z1^2)))
... disp("1/z2 = ", 1/z2, "|1/z2| = ", abs(1/z2), "arg(1/z2) = ", atand(imag(1/z2), real(1/z2)))
... disp("z1+2z2 = ", z1+2*z2, "|z1+2z2| = ", abs(z1+2*z2), "arg(z1+2*z2) = ", atand(imag(z1+2*z2), real(z1+2*z2)))
... disp("z1^5 = ", z1^5, "|z1^5| = ", abs(z1^5), "arg(z1^5) = ", atand(imag(z1^5), real(z1^5)))
...
endfunction
```

```
--> compl()

"z1+z2 = "

-1. + 4.i

"|z1+z2| = "

4.1231056

"arg(z1+z2) = "

104.03624

"3z1 = "

3. + 9.i

"|3z1| = "

9.4868330

"arg(3*z1) ="

71.565051

"z1/z2 = "

0.2 - 1.4i

"|z1/z2| = "

1.4142136

"arg(z1/z2) = "

-81.869898

"z1^2 = "

-8. + 6.i

"|z1^2| = "

10.

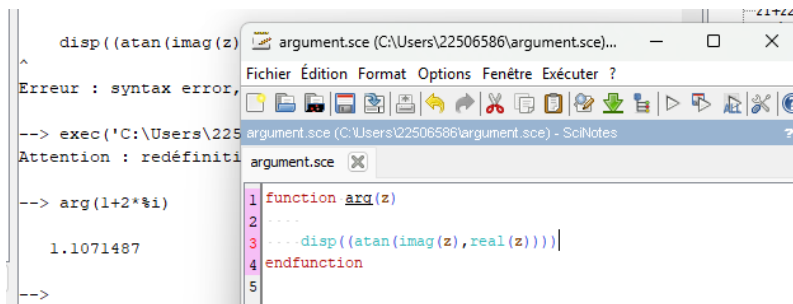
"arg(z1^2) ="

143.13010
```

DIAZENZA MOUANDA Regis Japhet
BALDE Mamadou

```
"1/z2 = "  
  
-0.4 - 0.2i  
  
"|1/z2| = "  
  
0.4472136  
  
"arg(1/z2) = "  
  
-153.43495  
  
"z1+2z2 = "  
  
-3. + 5.i  
  
"|z1+2z2| = "  
  
5.8309519  
  
"arg(z1+2*z2) = "  
  
120.96376  
  
"z1^5 = "  
  
316. - 12.i  
  
"|z1^5| = "  
  
316.22777  
  
"arg(z1^5) ="  
  
-2.1747441
```

Exercice 4 : Fonction argument(radian)

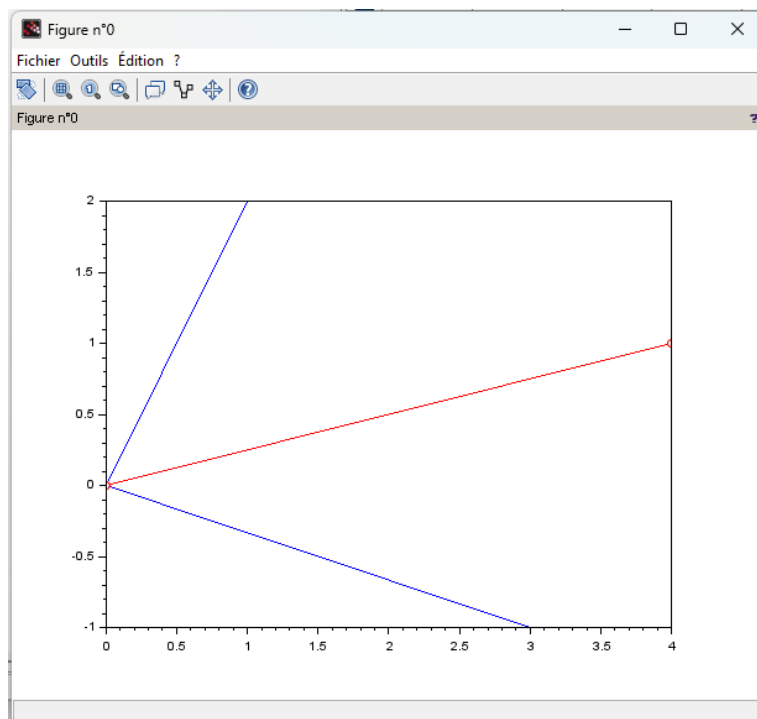


```
disp((atan(imag(z)/real(z))))  
^  
Erreur : syntax error, unexpected ')', expecting ')', at line 1, column 25.  
--> exec('C:\Users\22506586\argument.sce')  
Attention : redéfinition de la fonction 'arg' dans 'argument.sce'.  
  
--> arg(1+2*i)  
  
1.1071487  
  
-->
```

```
1 function arg(z)  
2 .....  
3 .....disp((atan(imag(z),real(z))))  
4 endfunction  
5
```

Exercice 5 : Représentation graphique des nombres complexes

```
--> z1=3-%i;  
--> z2=1+2*%i;
```



DIAZENZA MOUANDA Regis Japhet
BALDE Mamadou

```
--> plot([0,real(z1)], [0,imag(z1)])  
  
--> plot([0,real(z2)], [0,imag(z2)])  
  
--> plot([0,real(z1+z2)], [0,imag(z1+z2)], 'ro-')
```

Exercice 6 : Un peu d'électricité

Impédance complexe

$$i = 2\sqrt{2} \sin(314t + \frac{\pi}{6}) \quad u = 220\sqrt{2} \sin(314t)$$
$$\underline{I} = [2; \frac{\pi}{6}] = 2e^{\frac{\pi}{6}j}$$
$$\underline{U} = [220; 0] = 220e^0$$
$$\underline{Z} = \frac{\underline{U}}{\underline{I}} = \frac{220e^0}{2e^{\frac{\pi}{6}j}} = 110e^{-\frac{\pi}{6}j} = 110(\cos(-\frac{\pi}{6}) + j\sin(-\frac{\pi}{6}))$$
$$= 16,967 + 108,683j$$

```
--> i = 2*exp(30*i);  
  
--> u = 220*exp(0*i);  
  
--> z=u/i;  
  
--> z  
z =  
  
16.967659 + 108.683481i
```

Exercice 7 :

DIAZENZA MOUANDA Regis Japhet
BALDE Mamadou

```
--> zc=1/((100^-6)*2*pi*50*i);
```

```
--> zc
```

```
zc =
```

```
0. - 3.183D+09i
```

1- Calcul de la pulsation

$$\omega = 2\pi f \quad \text{avec } f = 50 \text{ Hz}$$

$$\omega = 2\pi \times 50 = 100\pi$$

2- Impédance complexe du résistor

$$u = ZI \quad \text{avec } Z = R$$

$$\underline{Z} = R = 30 \, \Omega$$

3- Impédance complexe du condensateur

$$\underline{Z} = \frac{1}{j\omega C} = \frac{1}{j \times 100 \times 10^{-6} \times 100\pi} = -31j$$

3- Impédance de la bobine

$$\underline{Z} = j\omega L = j \times 0,2 \times 100\pi = 20\pi j$$